

Modelling Fall 2025

5 ECTS, 2 x 2 hours a week

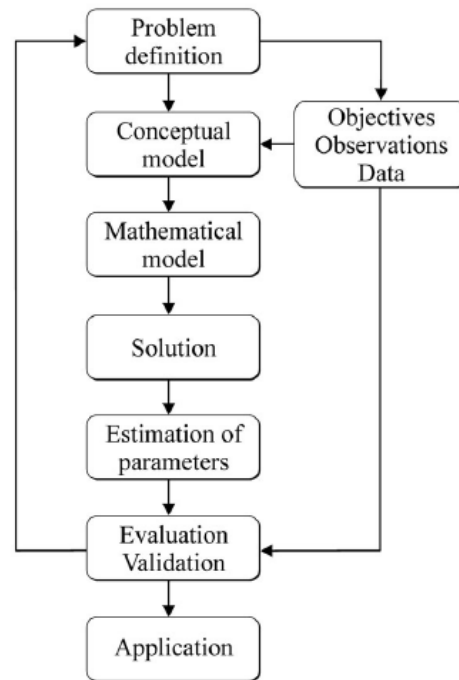


Figure 1.5. The different steps in model development.

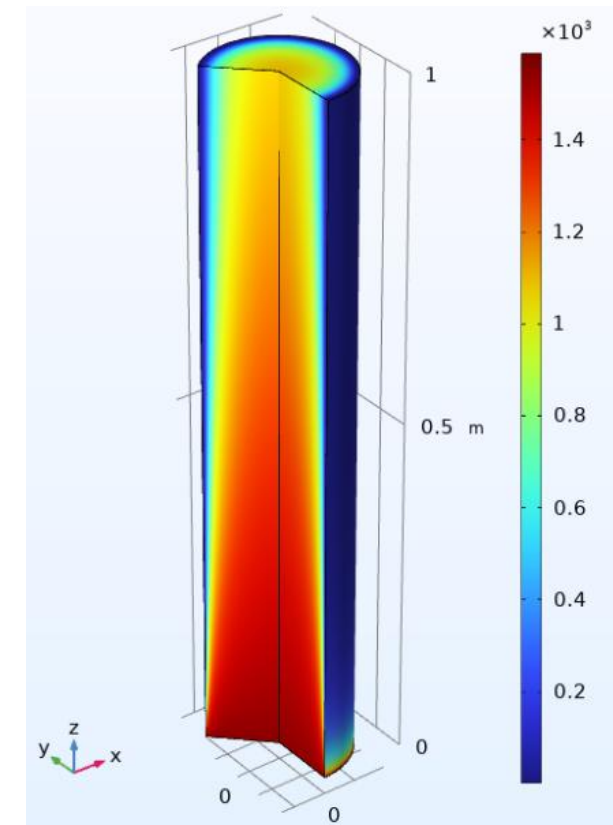
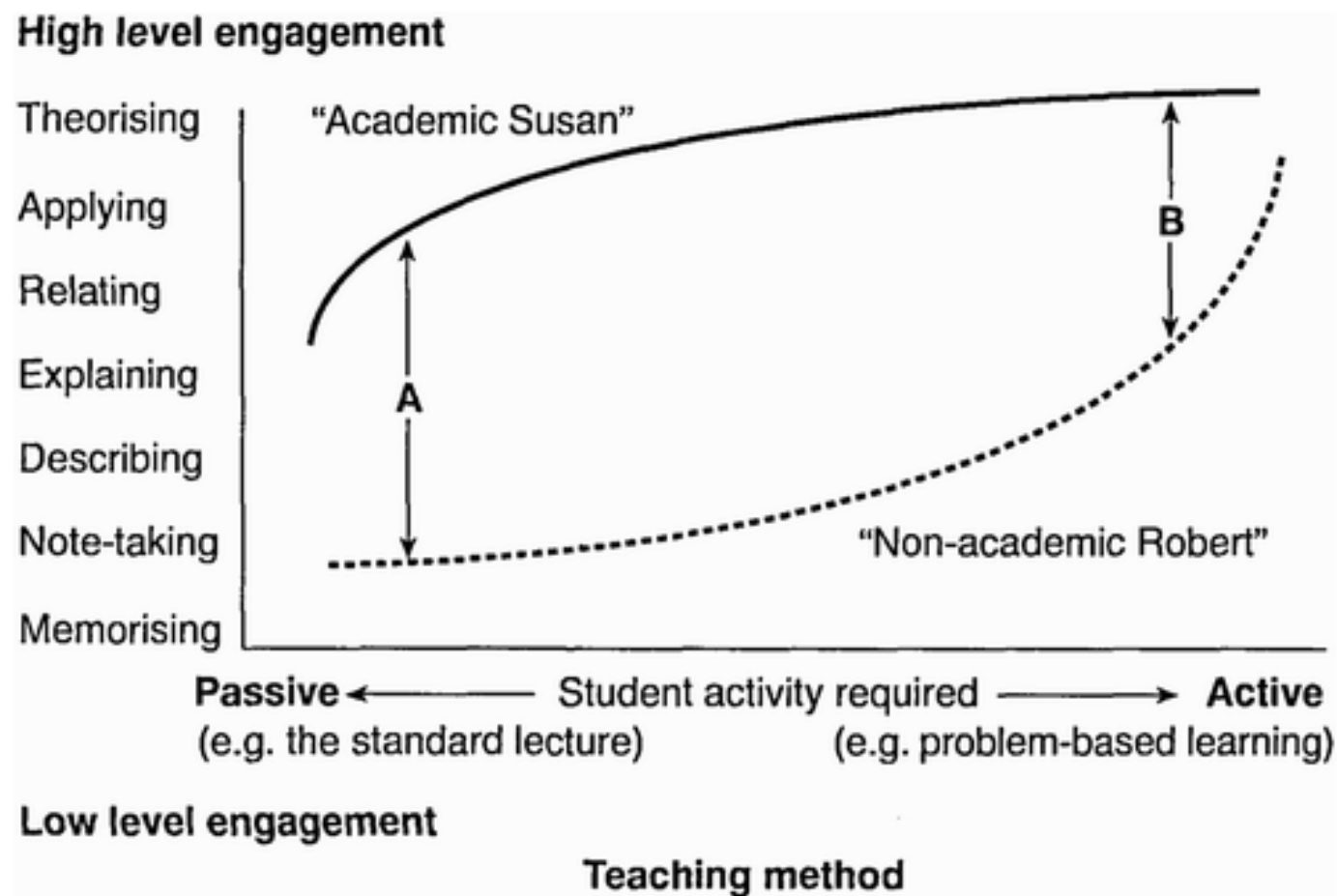
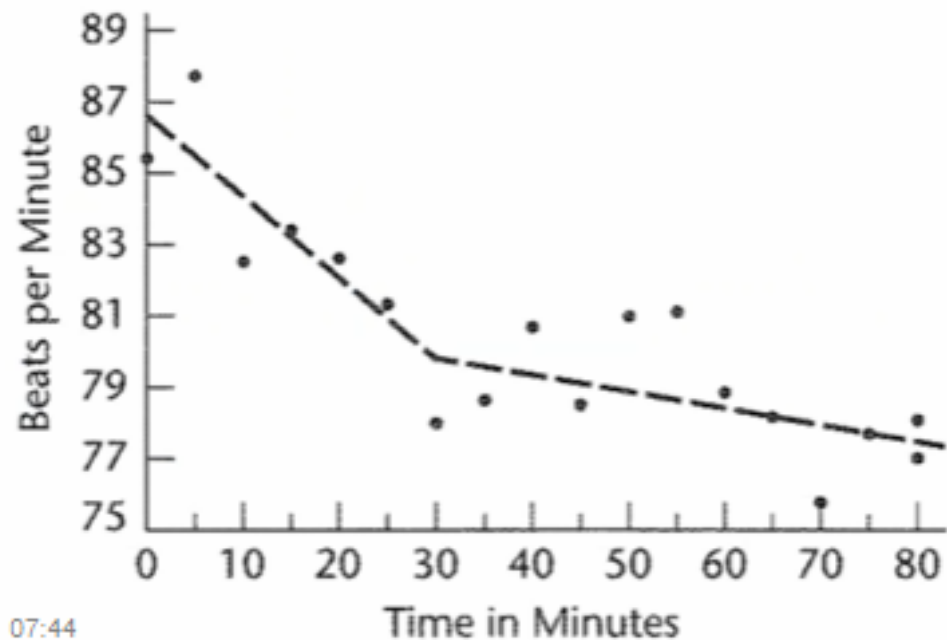


Figure 1. Student orientation, teaching method, and level of engagement.

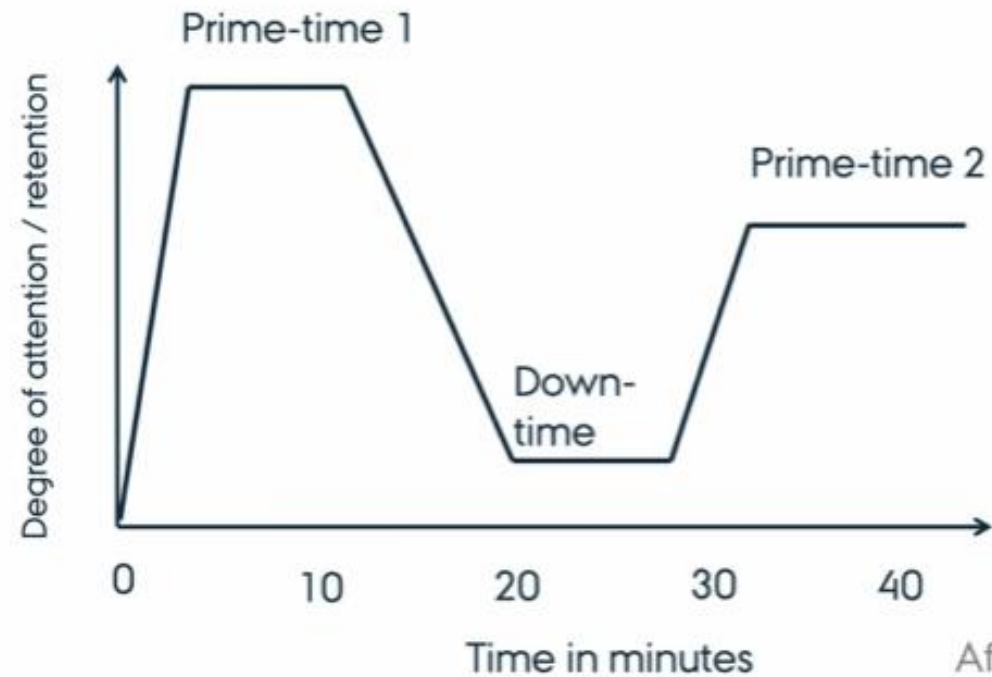


**FIGURE 3.6. STUDENTS' HEART RATES
IN UNINTERRUPTED LECTURES.**



07:44

The retention-attention curve



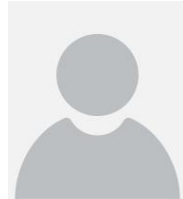
After Sousa 2001

Instructors and teachers

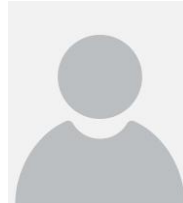
- Sasha Hafner (Senior researcher BCE Environmental Engineering)



- Rasmus (MSc. student)



- Mikkel (MSc. student)



- Anne (research assistant)



- Frederik Dalby (Tenure track assistant professor, BCE Environmental Engineering)



Lecture plan

Date	Location	Class	Course Week	Year week	Day	Time	Teacher	Reading	Topic
26-08-2025	1170-347	1 1		35	Tuesday	12:00-14:00	Frederik	Rasmuson Chapters 1 & 2	Introduction to course, purpose of models, model classification, the modelling procedure
28-08-2025	5220-318	2 1		35	Thursday	08:00:10:00	Sasha	Rasmuson Chapter 3	Model formulation
02-09-2025	1170-347	3 2		36	Tuesday	12:00-14:00	Sasha	Incropera Sections 1.1-1.2, py_data.md	Heat transfer and analytical models in Python
04-09-2025	5220-318	4 2		36	Thursday	08:00:10:00	Sasha	Cussler Chapter 1	Mass transfer and finite difference method
09-09-2025	1170-347	5 3		37	Tuesday	12:00-14:00	Sasha	Cussler Section 8.5, Kong Section 22.6	Interphase interface mass transfer and solve_ivp()
11-09-2025	5220-318	6 3		37	Thursday	08:00:10:00	Frederik		Mini-project
16-09-2025	1170-347	7 4		38	Tuesday	12:00-14:00	Frederik		Mini-project
18-09-2025	5220-318	8 4		38	Thursday	08:00:10:00	Frederik	Edsberg notes, Python book chap 20/Snape p 219	PDEs and MOL 1D directlet
23-09-2025	5344-101	9 5		39	Tuesday	12:00-14:00	Frederik	Incropera BCs/Edsberg notes	Neumann/robin 1D and 2D
25-09-2025	5220-318	10 5		39	Thursday	08:00:10:00	Frederik	Edsberg notes	Stability of ODEs
30-09-2025	1170-347	11 6		40	Tuesday	12:00-14:00	Frederik		Coupling mass balance and temp dependent reaction dynamics
02-10-2025	5220-318	12 6		40	Thursday	08:00:10:00	Frederik		1D and 2D with heat, mass, reaction (reaction temp dependent and or biological reaction)
07-10-2025	5344-101	13 7		41	Tuesday	12:00-14:00	Frederik	Chapter 4	Empirical models/dimension analysis
09-10-2025	5220-318	14 7		41	Thursday	08:00:10:00	Frederik	Chapter 4	Empirical models/dimension analysis
14-10-2025		BREAK		42	Tuesday				
16-10-2025		BREAK		42	Thursday				
21-10-2025	5344-101	15 8		43	Tuesday	12:00-14:00	Sasha		Parameter estimation
23-10-2025	5220-318	16 8		43	Thursday	08:00:10:00	Sasha		Parameter estimation
28-10-2025	5344-101	17 9		44	Tuesday	12:00-14:00	Sasha		Model complexity and simplification
30-10-2025	5220-318	18 9		44	Thursday	08:00:10:00	Sasha		Model complexity and simplification
04-11-2025	5344-101	19 10		45	Tuesday	12:00-14:00	Frederik		Mini-project
06-11-2025	5220-318	20 10		45	Thursday	08:00:10:00	Frederik		Mini-project
11-11-2025	5344-101	21 11		46	Tuesday	12:00-14:00	Frederik		Model evaluation and validation
13-11-2025	5220-318	22 11		46	Thursday	08:00:10:00	Frederik		Model evaluation and validation
18-11-2025	5344-101	23 12		47	Tuesday	12:00-14:00	Frederik		Buffer
20-11-2025	5220-318	24 12		47	Thursday	08:00:10:00	Frederik		Buffer
25-11-2025	5344-101	25 13		48	Tuesday	12:00-14:00	Frederik		Project
27-11-2025	5220-318	26 13		48	Thursday	08:00:10:00	Frederik		Project
02-12-2025	5344-101	27 14		49	Tuesday	12:00-14:00	Frederik		Evaluation + Model project report
04-12-2025	5220-318	28 14		49	Thursday	08:00:10:00	Frederik		Evaluation + Model project report

<https://github.com/AU-BCE-EE/course-modelling-2025/blob/main/schedule.csv>

Search titles, descriptions



Welcome to modelling

Week1

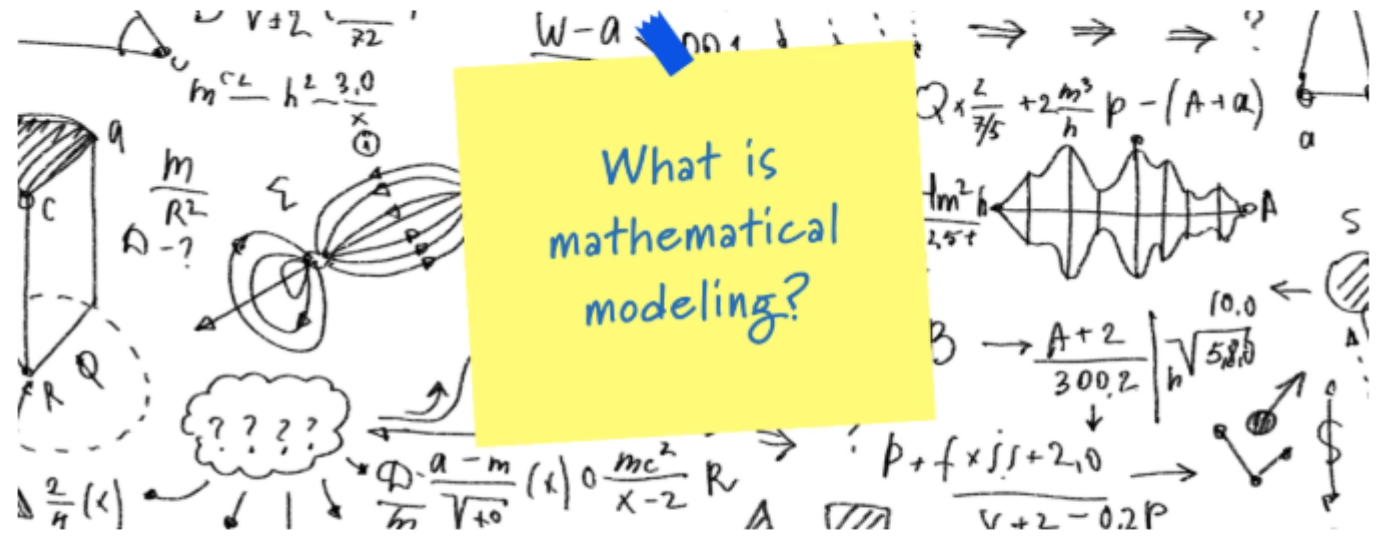
Week2

Week3

Week4

Week5

Welcome to modelling



Litterature: Parts of Rasmuson et al. 2014. Mathematical Modelling in Chemical Engineering.

Other notes will be shared.

Course description: <https://kursuskatalog.au.dk/da/course/134371/Modellering>

See Github for more information about the course and files

Github link: <https://github.com/AU-BCE-EE/course-modelling-2025>

Search titles, descriptions



Welcome to modelling

Week1

Week2

Week1

Tuesday: Introduction to modelling course, modelling procedure and classification

Readings: Chap 1 & 2 in Rasmuson et al.

Link: <https://github.com/AU-BCE-EE/course-modelling-2025>

Thursday: Model formulation and conservation and rate equations

Readings: Chap 3 in Rasmuson et al.

Link: <https://github.com/AU-BCE-EE/course-modelling-2025>

Learning Outcomes

Participating students will after this course be able to:

- Describe the concepts behind different mathematical approaches to acquire numerical solutions
- Criticize models for their applicability in solving a specific problem
- Use computer code to create a mathematical model for a given problem and reflect on its results
- Perform parameter estimations and compare system performance
- Analyse through validation the quality of model and its predictive capabilities

Examination

FORM OF EXAMINATION

Oral

FORM OF CO-EXAMINATION

Internal co-examination

ASSESSMENT

7-point grading scale

PERMITTED EXAM AIDS

All

DURATION

25 minute(s)

Requirements for taking the exam

Submission of a written group report at the end of the semester. There is a maximum of 5 students in each group and all team members are responsible for the entire written report. The written report must be submitted on time and must be submitted as a single file in pdf format.

Comments

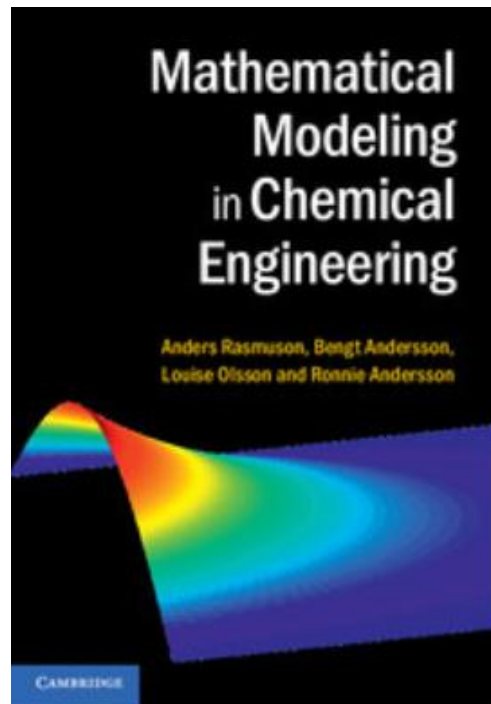
Oral exam: The course is evaluated by an oral exam, which is held on the basis of the submitted group report and the learning objectives of the course.

Assessment: The exam is assessed as an overall assessment of the individual oral examination.

[READ MORE](#) ▾

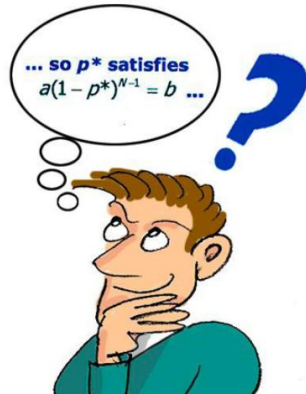
Resources

Litterature: Rasmuson et al. 2014, notes and notes from other books. Also computer algorithm (python) book for refreshing ODE and BVP



”A mathematical model is an **abstract description** of a **concrete system** using **mathematical concepts and language**”, wikipedia

What is a mathematical model?



“A mathematical model is a **representation** of a **real-world system** using **mathematical language** and structures such as equations, functions, variables, and parameters”, ChatGPT

“All models are **abstractions** of **real systems** and processes. Nevertheless, they serve as tools for engineers and scientists to develop an understanding of important systems and processes using **mathematical equations**” Rasmuson et al. 2014

”All models are wrong but some are useful”

But what is mathematical modelling?

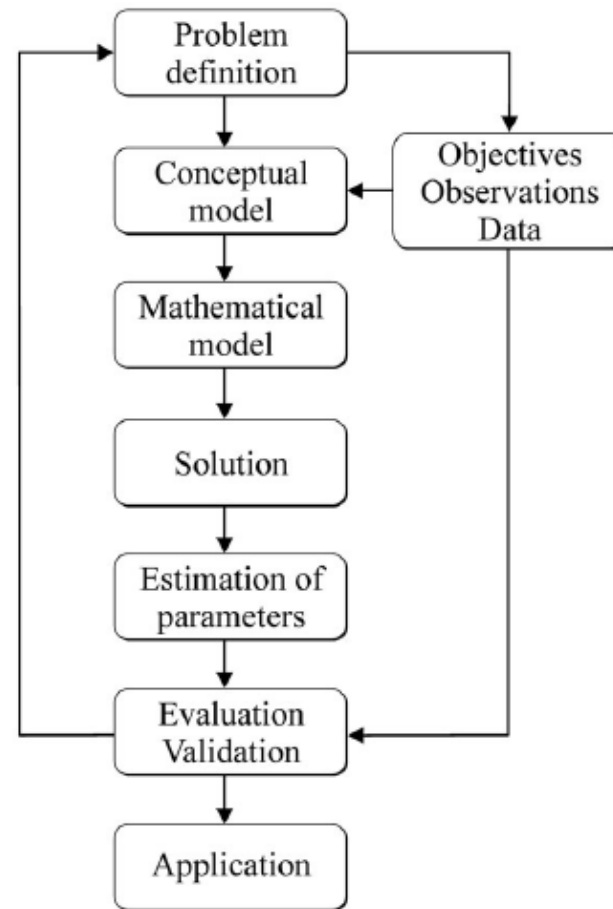
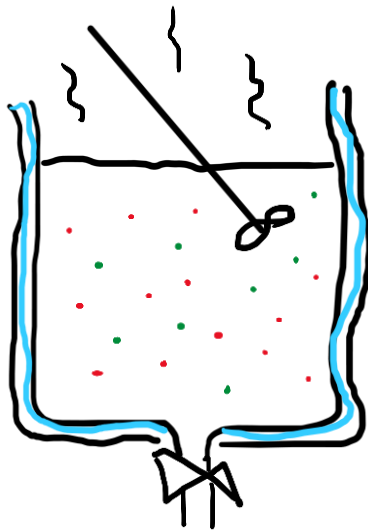


Figure 1.5. The different steps in model development.

An open fermentor, which operates in batch mode and contains a solution of water with two reactants that react to produce some product. The reactor is cooled on the outside and mixed with a stirrer



model it!!!

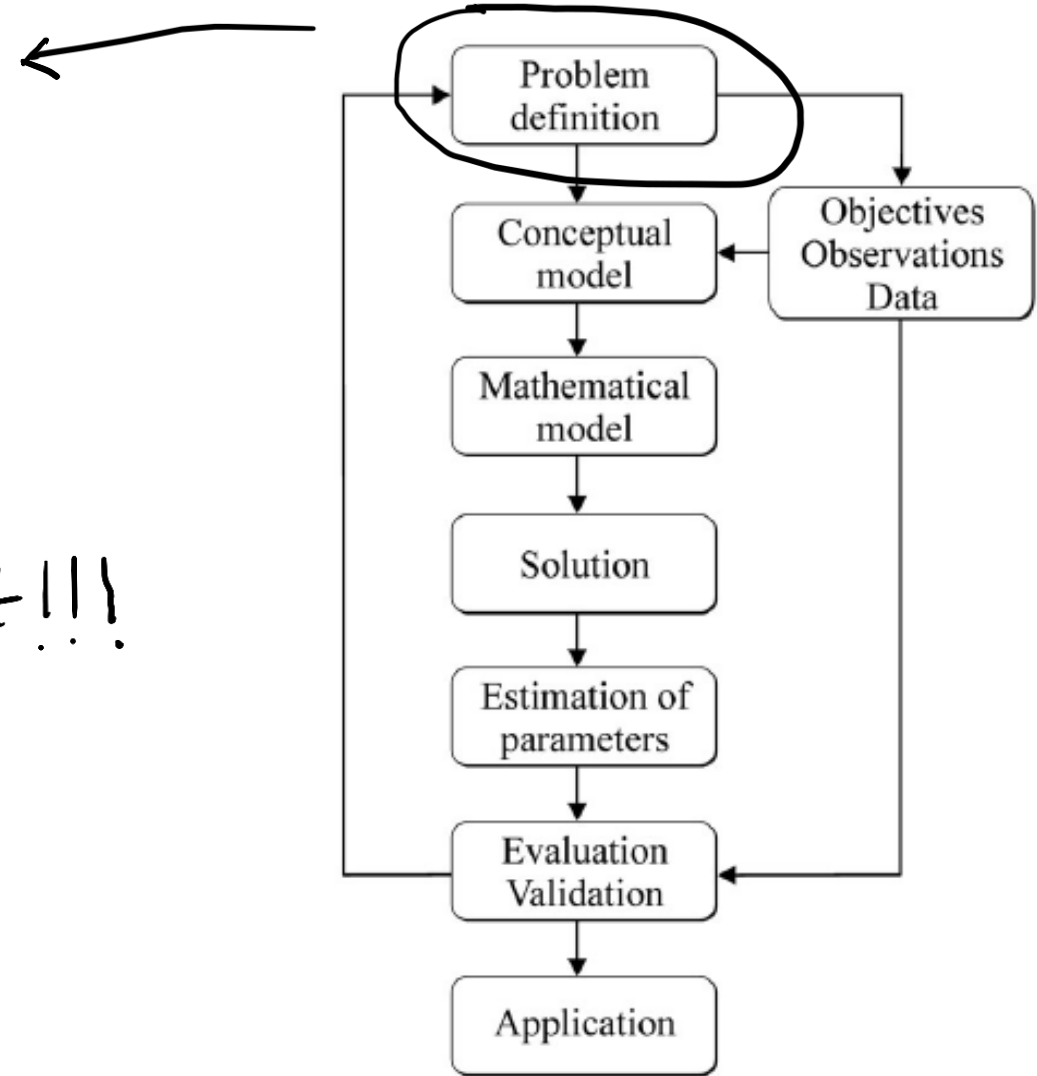
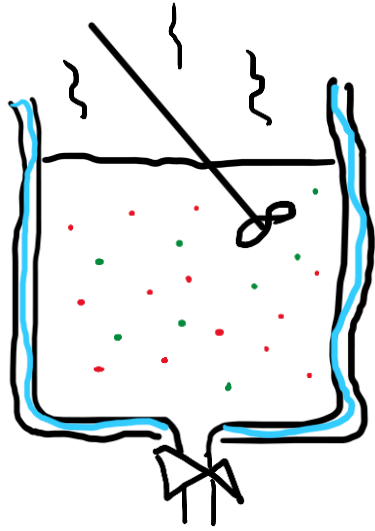


Figure 1.5. The different steps in model development.



1. How much heat should be removed from the system to maintain a temperature of $20\text{ }^{\circ}\text{C}$?
2. How long time does it take to convert 90% of reactant A?
3. Will the reactor dry out due to water evaporation?
4. At what speed should the mixer spin to prevent particles from settling?
5. Etc...

Is the reaction exothermic? $\rightarrow$ propably, since it is a fermentation reaction, which is normally exothermic

Is heat production constant? $\rightarrow$ No because it is a batch system

Does the question make sense then? $\rightarrow$ No. So we need more accurate problem statement

Problem definition: What should the cooling flowrate be through the cooling jacket over time and until 90% of reactant A is converted from the system to maintain a constant temperature of $20\text{ }^{\circ}\text{C}$?

What do we know about the system?
Do we have data?
Do we need to make **assumptions**?
Identify physical/chemical principles of relevance?

assumptions?
Heat :
mixer : + heat
reaction : + heat
evaporation : ÷ heat
jacket : ÷ heat
data ?

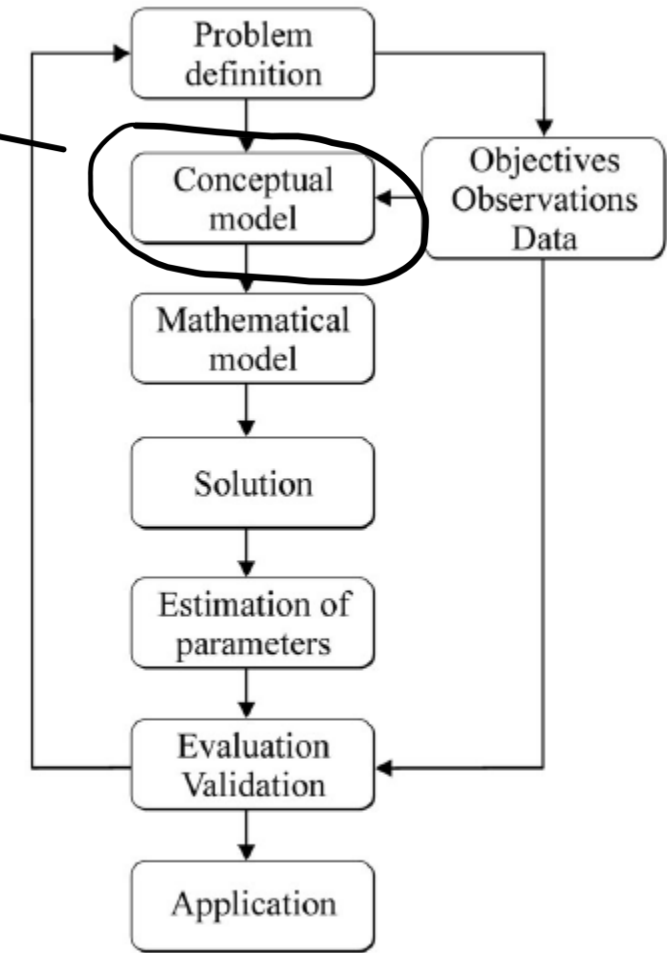
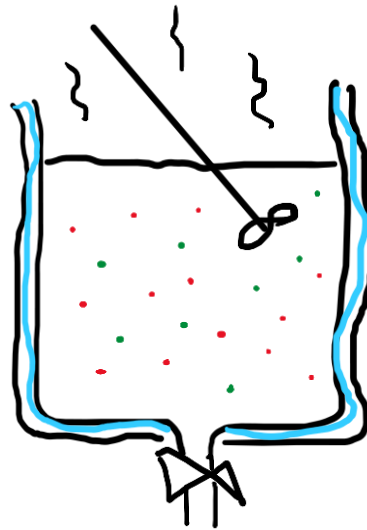
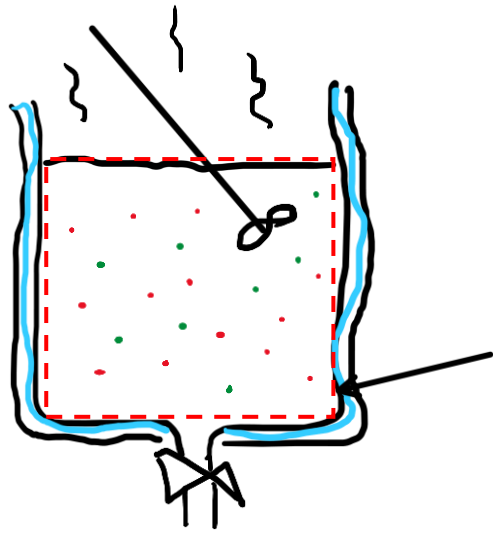


Figure 1.5. The different steps in model development.



Independent variable → time
 Dependent variable → Many! (but we are interested in "heat removed" from the system (which depends on other time dependent variables)
 Or are we interested in temperature of the system?
 Boundary of system to model?

Mass balance

$$\left[\begin{array}{c} \text{accumulation} \\ \text{within system} \end{array} \right] = \left[\begin{array}{c} \text{net transport in} \\ \text{through system} \\ \text{boundaries} \end{array} \right] + \left[\begin{array}{c} \text{net generation} \\ \text{within system} \end{array} \right]$$

Energy balance

$$\left[\begin{array}{c} \text{accumulation} \\ \text{of TE} \\ \text{within system} \end{array} \right] = \left[\begin{array}{c} \text{net flow of } U, PKE, \\ PE, KE \text{ through} \\ \text{system boundaries} \end{array} \right] = \left[\begin{array}{c} \text{net generation} \\ \text{within} \\ \text{system} \end{array} \right]$$

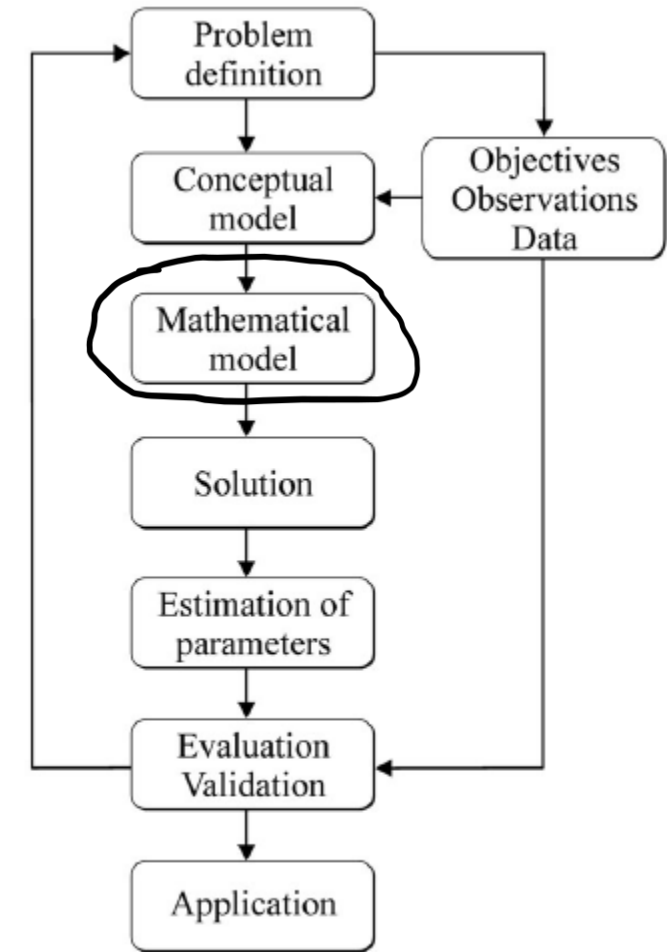
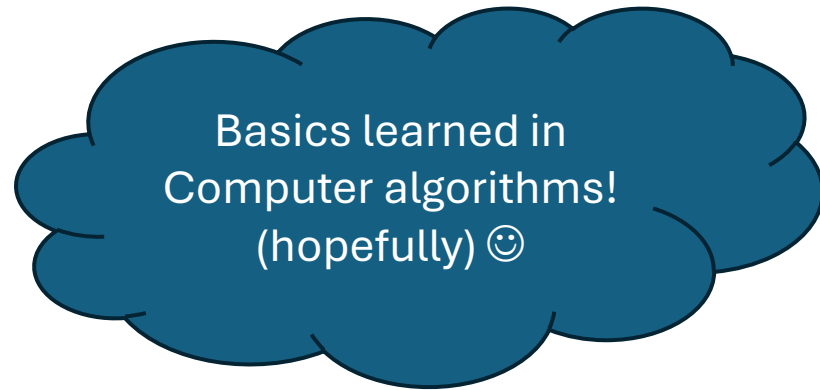


Figure 1.5. The different steps in model development.



Additional solution theory in this course

- Implement different BCs
- Implement in higher dimensions
- Stability of model and solution method

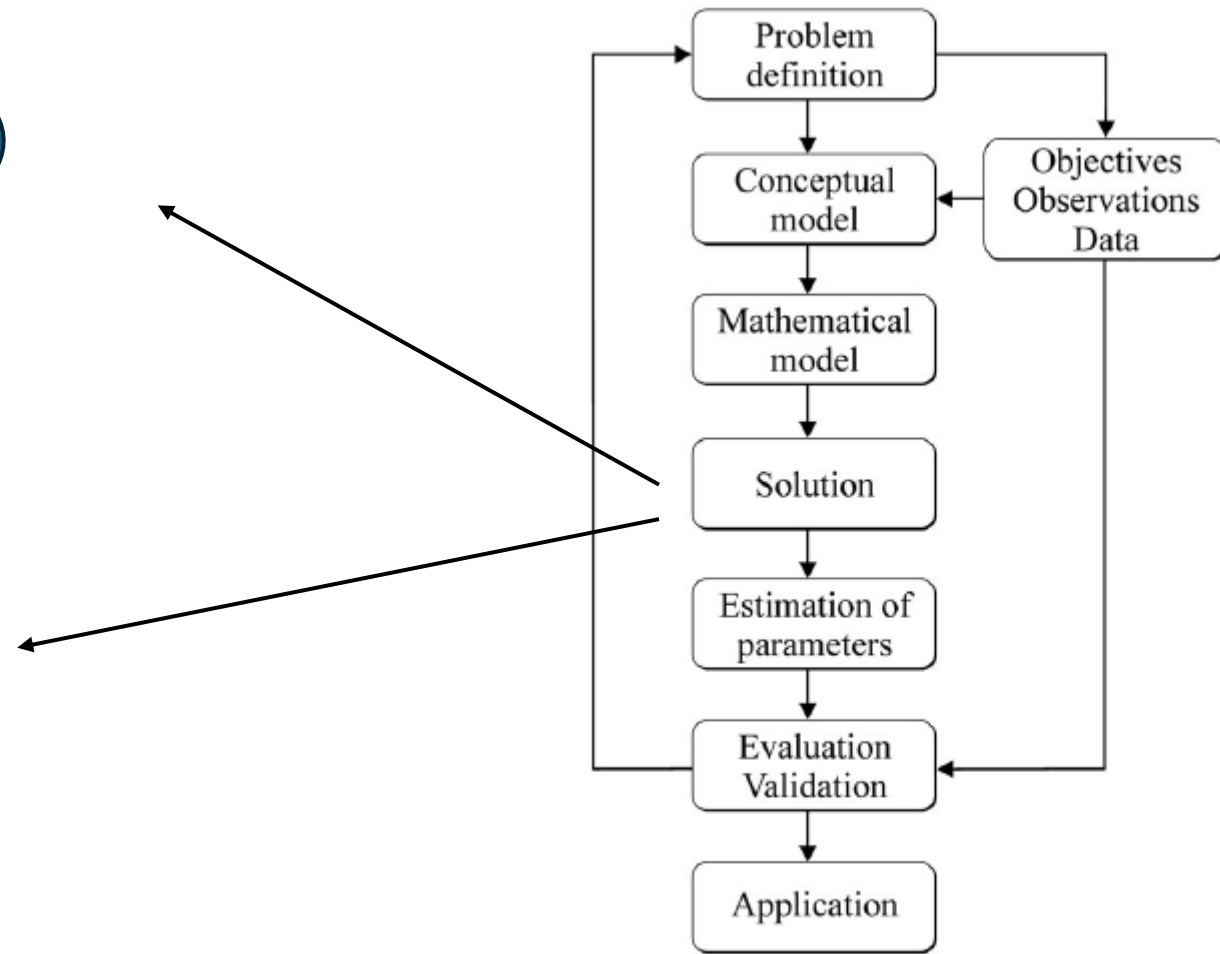
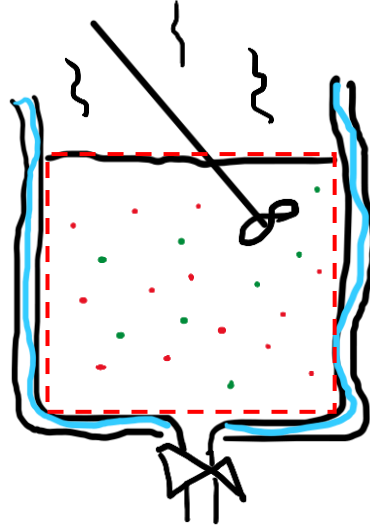


Figure 1.5. The different steps in model development.

Need data!



Unknown parameters?

Rate constants: $k * A * B$

Heat transfer coefficients: $Q = h * A * (T_2 - T_1)$

Known parameters?

Size of tank, cooling fluid flow rate,
Surface area of cooling jacket etc.

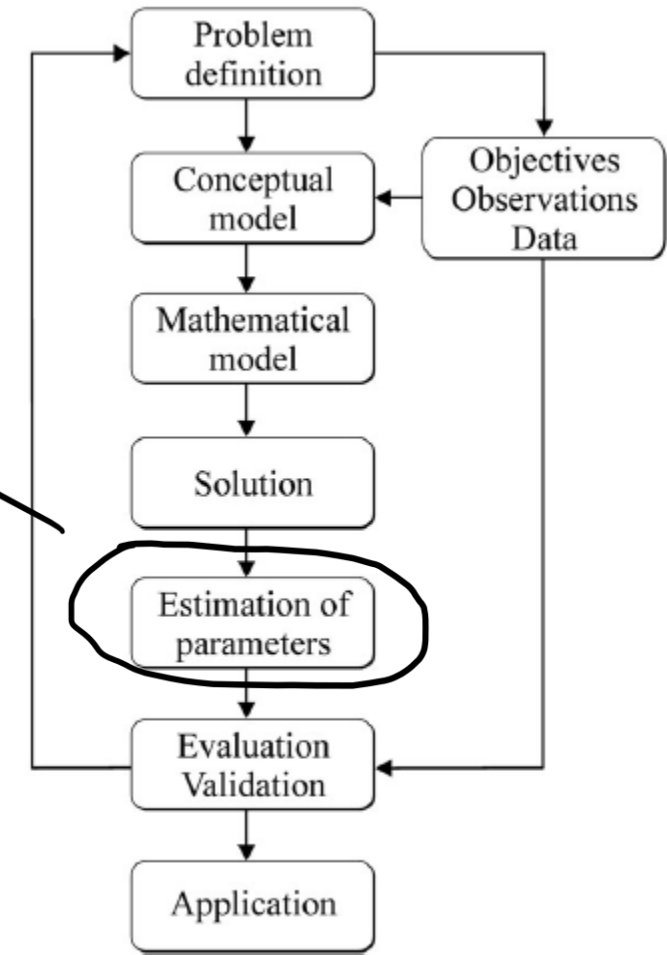


Figure 1.5. The different steps in model development.

- accuracy (is the output of the model correct?);
- descriptive realism (i.e. based on correct assumptions);
- precision (are predictions in the form of definite numbers?);
- robustness (i.e. relatively immune to errors in the input data);
- generality (applicable to a wide variety of situations);
- fruitfulness (a model is considered fruitful if its conclusions are useful or if it inspires development of other good models).

Reformulate model in iterative approach

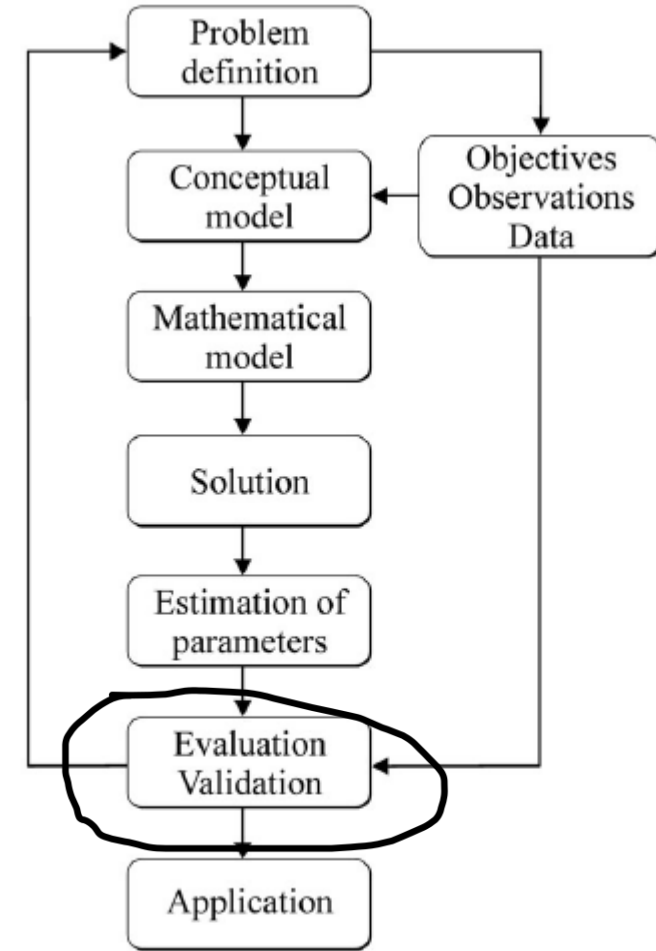


Figure 1.5. The different steps in model development.

How can we quantify accuracy with metrics?

Mean squared error (MSE)

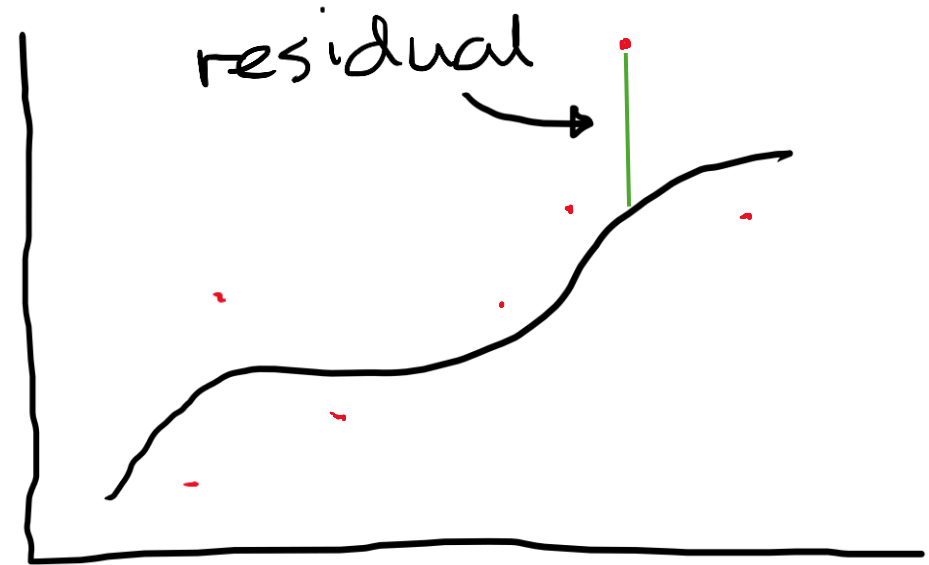
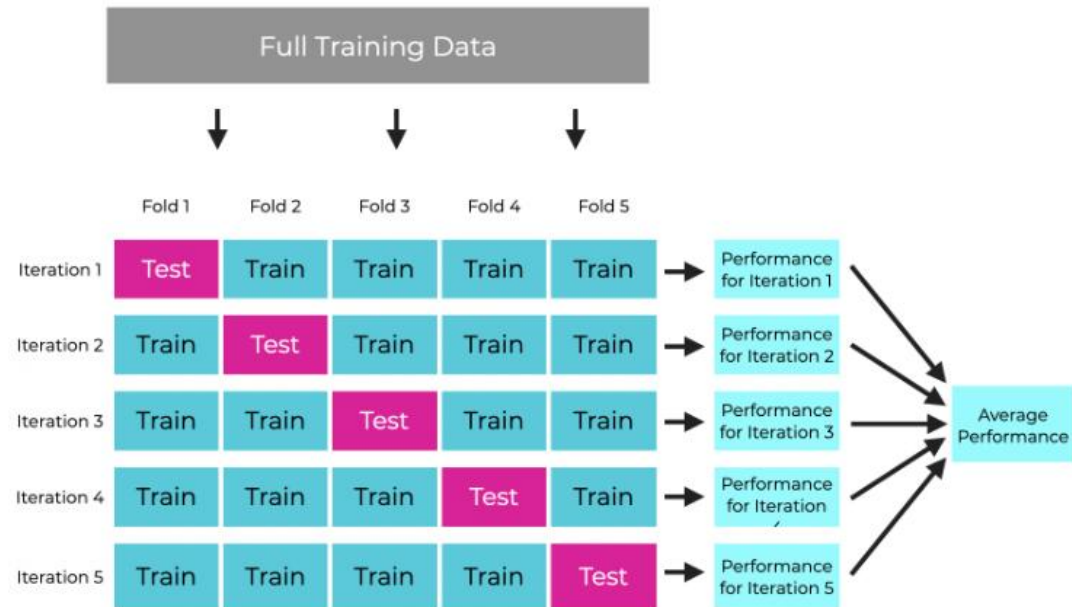
Root mean squared error (RMSE)

Mean absolute error (MAE)

R-squared or adjusted R-squared (R^2)

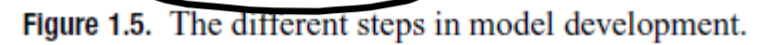
How can we quantify generality?

Leave-out part of data approach (cross validation)

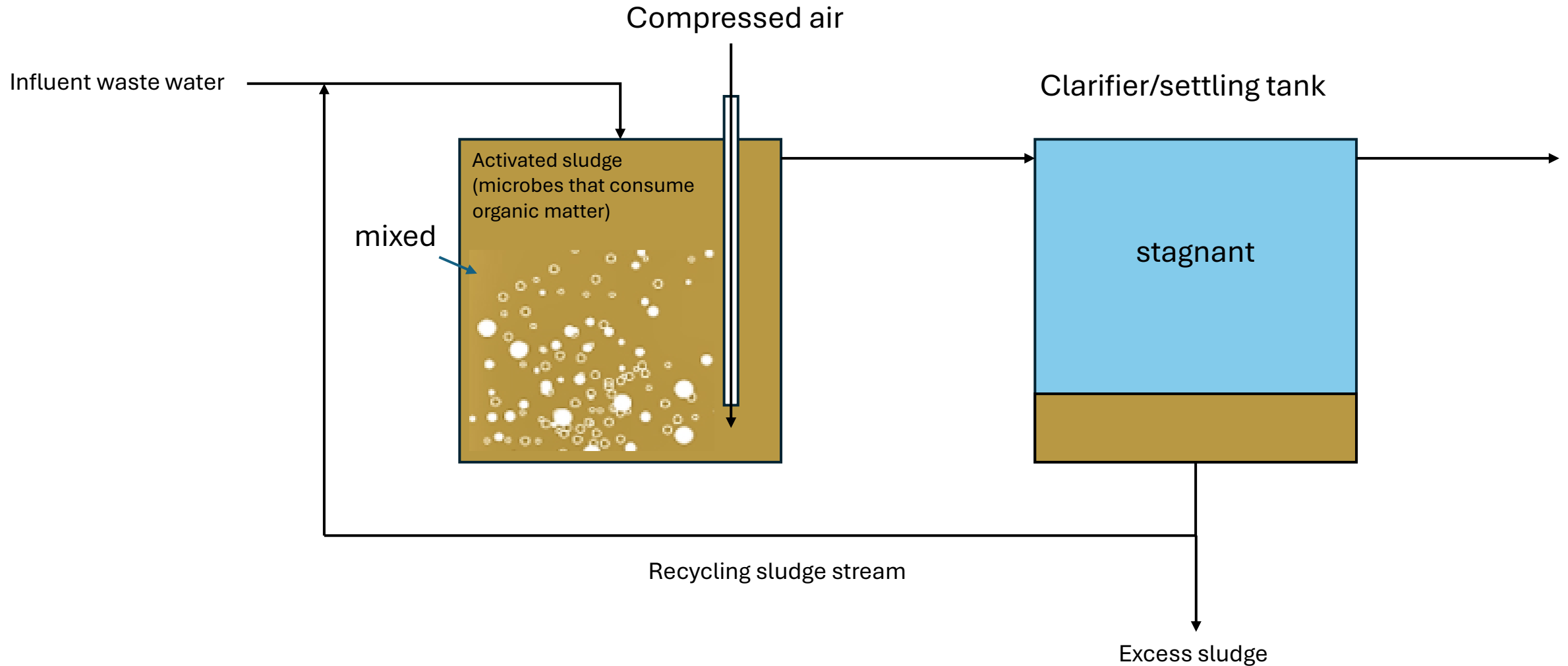


1. Save money
2. Reduce resources
3. Increase knowledge
4. Improve your system efficiency
5. Return to model scheme if the system is changed

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Waste water model application (example)



Use to control recycling stream and compressed air flow
(saves money for energy and ensures clean stream out of clarifier)

Discussion exercises

1. Read example 1.3. in Rasmuson et al.

Discuss with a classmate, how and what elements in the modelling process you can recognize from the modelling process scheme?

2. Discuss questions below with a classmate

- (1) Give some reasons for doing mathematical modeling in chemical engineering.
- (2) Explain why the model development often becomes an iterative procedure.